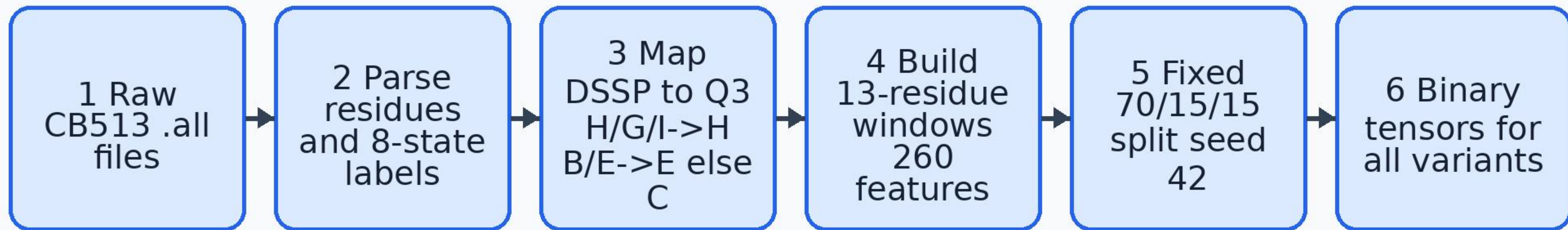
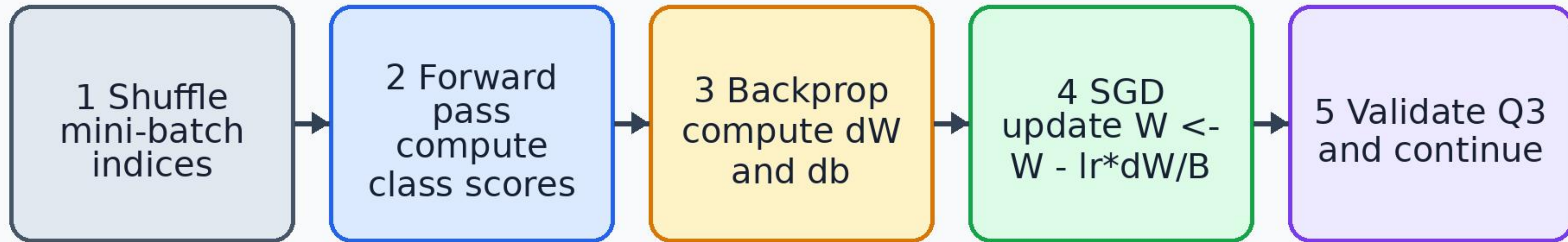


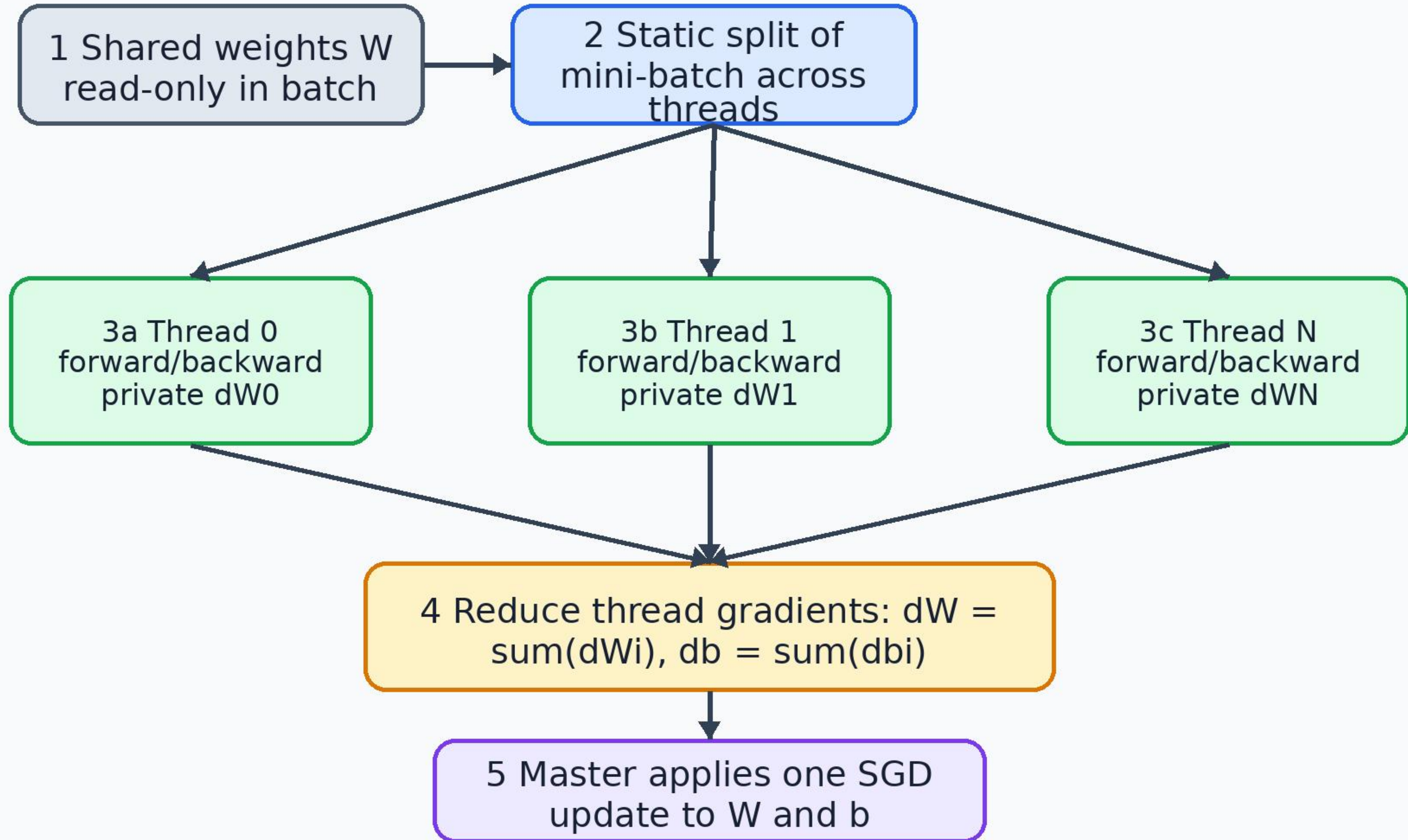
CB513 Preprocessing



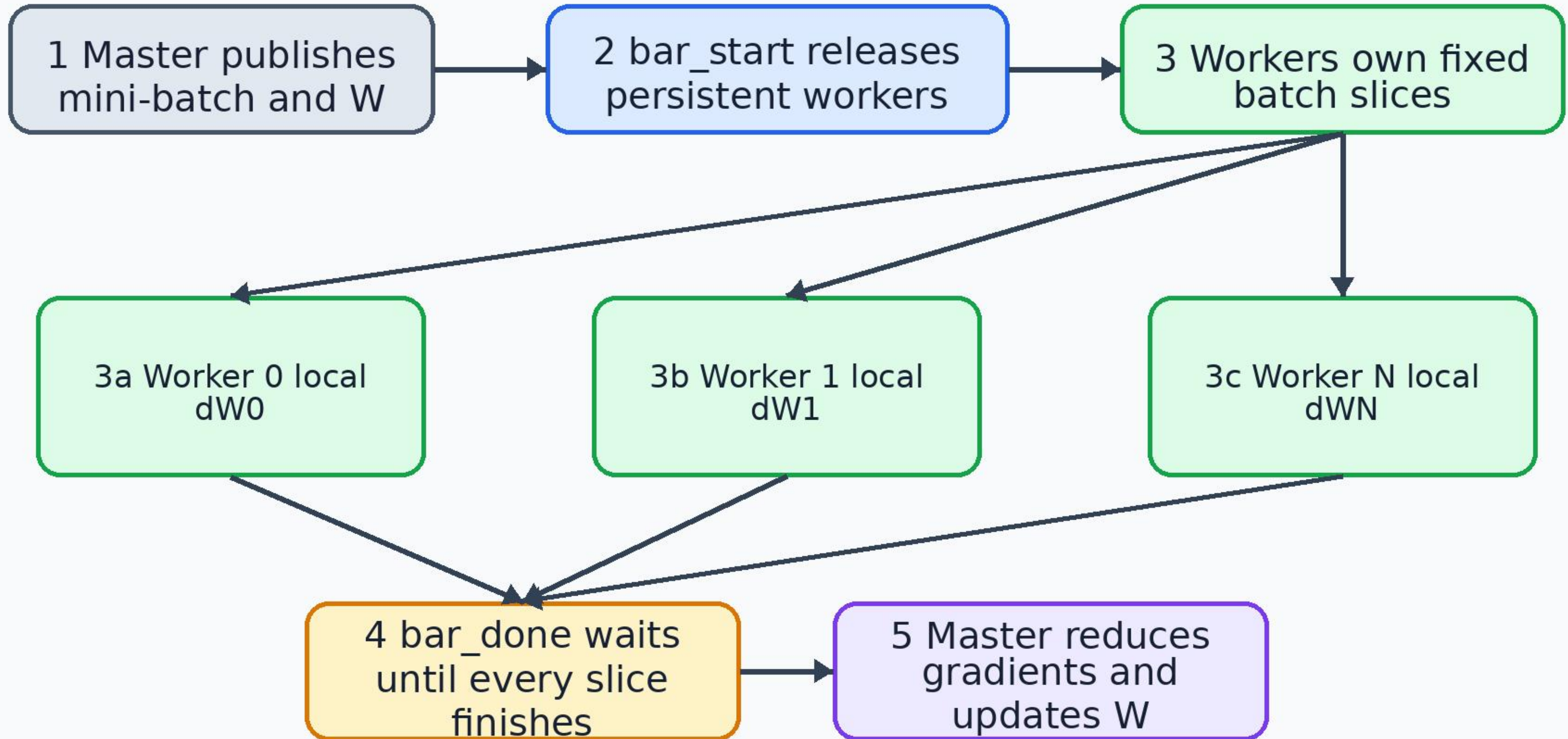
Serial Mini-batch SGD



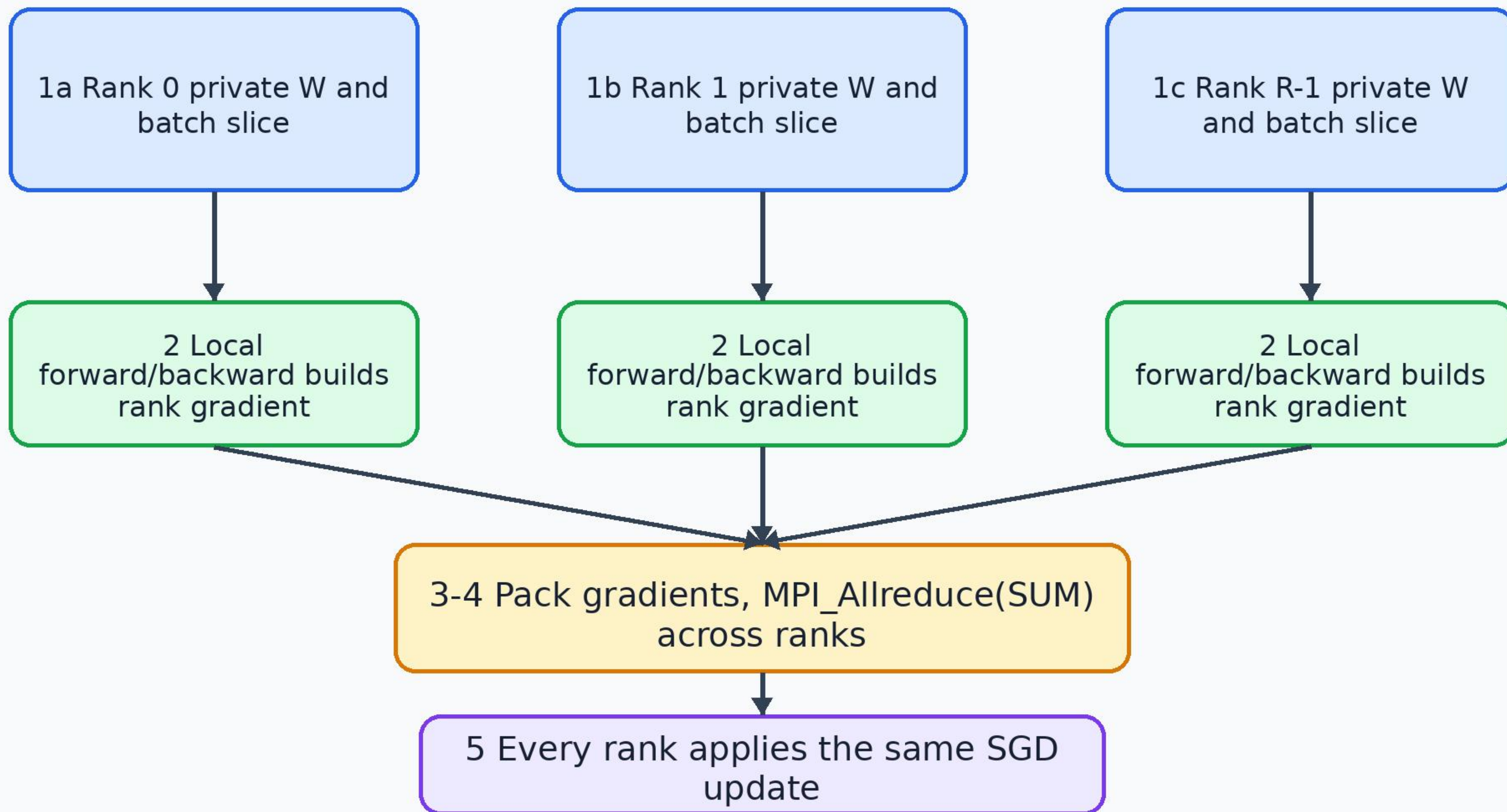
OpenMP Shared-memory Data Parallelism



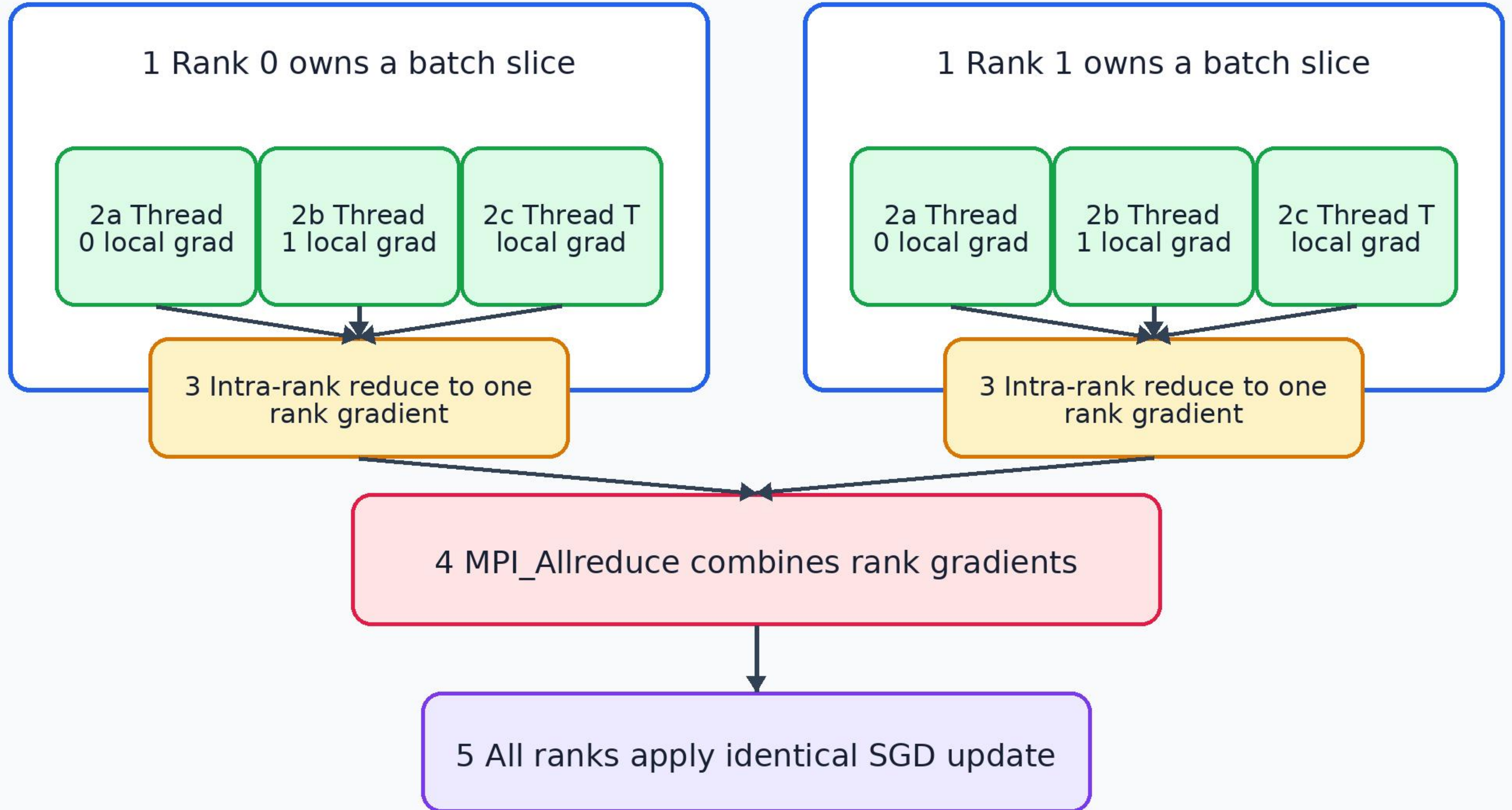
POSIX Threads Master / Worker Pool



MPI Distributed-memory Training



Hybrid MPI + OpenMP



CUDA GPU Training Workflow

CPU host

1 Shuffle batch indices and launch kernels

GPU device

2 Data and W stay resident in GPU memory

3 Parallel gather builds column-major batch

4 Forward GEMMs plus ReLU kernels

5 Softmax loss computes dZ in parallel

6 Backward GEMMs update W in place

7 Copy W to CPU only for Q3 evaluation

